Numerical model of hydrodynamic instabilities

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1 Introduction

Instabilities shape a large part of the Universe. They influence the evolution of gas clouds into galaxies, stars and planets, along with the evolution of more impulsive events such as supernovae or jets (Zhou et al., 2021). On smaller scales, the same instabilities influence Earth, such as waves in the oceans and effects in the magnetosphere and ionosphere. While most of them can be sorted into a few conceptually simple classes, the fluid dynamics they create are not as straightforward.

Trussoni (2008) defines an instability as a system's reaction to an external perturbation which definitely modifies its structure through an energy transfer. This is the case for both Kelvin-Helmholtz (KHI) and Rayleigh-Taylor instabilities (RTI). In the case of RTI, a denser fluid and a less dense fluid share an interface. The external element which introduces the perturbation are gravitational forces, which push the denser fluid towards the less dense fluid. It caves the interface towards the denser fluid in some parts and the less dense fluid in others, forming a characteristic mushroom shape (Zhou et al., 2021). If the discrepancy in densities is high enough, then the lower density fluid forms bubbles in the higher density fluid instead (Zhou et al., 2021). Richtmyer-Meshkov instabilities (RMI) are seen as a particular case of Rayleigh-Taylor instabilities, in which the perturbations start due to a shock-wave instead of a constant or mostly constant gravitational force. Because of this, the system is only unstable to RTI if the gravitational force is in the right direction, placing the denser fluid "above", while the system is always unstable to RMI regardless of the direction of the shock-wave (Zhou et al., 2021). As we will see in the Discussion, there seem to be some additional conditions necessary to produce RM instabilities. They are both observed in the evolution of supernovae, and in smaller scale events such as the flow of magma (Zhou et al., 2021).

KHI instabilities appear between two layers of material moving at different velocities (Zhou et al., 2021). Due to this motion, the interface between the two materials is unstable, creating instabilities (Matsumoto & Hoshino, 2004). These are also present in many phenomena, such as jets in the interstellar medium, or waves in the ocean. There often is a density difference between the two layers, which creates further instabilities (Matsumoto & Hoshino, 2004).

Both types of instabilities, KHI and RTI, are often observed together. The relative velocity of the two fluids moving in RT instabilities generates roll-ups which are due to KH instabilities (Zhou et al., 2021). As mentioned above, KH instabilities often happen between two material of different densities, which also produce RT instabilities (Matsumoto & Hoshino, 2004).

The growth of these instabilities start linearly, and can be modelled with simple exponential expansions (Zhou et al., 2021), however, past a certain point, the growth becomes non-linear and can only be modelled with numerical simulations. This non-linear stage happens in two parts, first a "weak" non-linearity during which the interface is still smooth, followed by a "strong" non-linearity during which the characteristic shapes of the different instabilities form (Allen & Hughes, 1984; Zhou et al., 2021). As those instabilities are common in many fields, understanding them is important to understand many phenomena. And as they can not be modelled analytically past a certain point, numerical simulations are necessary for this.

In this report, we will use a HLLE Riemann solver to model both Rayleigh-Taylor and Kelvin-Helmholtz instabilities in 2D, and attempt to model Richtmyer-Meshkov instabilities.

2 Method

2.1 Hydrodynamics

To model the evolution of fluids in the absence of a magnetic field or viscosity, the first important points are the Euler equations and the equation of state of the system, which govern how the system evolve in time and space. Here, we assume an adiabatic equation of state: $P = K\rho^{\gamma}$, where gamma is the adiabatic index. The choice of an adiabatic or isothermal equation of state can be important for some instabilities in compressible fluids such as we have here, but the effect is negligeable for RTI and KHI instabilities (Allen & Hughes, 1984).

There are three Euler equations, the continuity equation (eq. 1), which deals with the density displacement,

$$\frac{\partial \rho}{\partial t} + \partial_i(\rho u_i) = 0, \tag{1}$$

the momentum equation (eq. 2), which deals with the momentum changes,

$$\partial_t(\rho u_j) + \partial_i \rho u_i u_j + \delta_{ij} P = \rho g_j, \tag{2}$$

and finally, the energy equation (eq. 3), which as its name indicates deals with the change in energy.

$$\frac{\partial E}{\partial t} + \partial_i[(E + P)u_i] = 0. \tag{3}$$

This set of equations can be written in a more compact form (O'Connor, 2026), which makes its use in numerical simulations slightly easier. This is equation 4,

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F_i}(\mathbf{U})}{\partial x_i} = \mathbf{S}(\mathbf{U}), \tag{4}$$

in which $\mathbf{U}$ are the conserved variables, $\mathbf{U} = [\rho, \rho u_j, E]$, $\mathbf{F_i}(\mathbf{U})$ are the fluxes, $\mathbf{F_i}(\mathbf{U}) = [\rho u_i, \rho u_i u_j + \delta_{ij} P, (E + P)u_i]$, and $\mathbf{S}$ are the additional forces such as viscosity or gravity which could impact the fluid. Here, this is only the gravity in the RTI case, so $\mathbf{S}(\mathbf{U}) = [0, \rho g_j, 0]$.

In addition to these equations, the sound speed (eq. 5),

$$c_s = \gamma P/\rho, \tag{5}$$

is needed to determine the wave speed, which impacts the evolution of the simulations, and the time-step. The pressure of the fluid is also derived from numerically evolved quantities (eq. 6),

$$P = \rho\epsilon(\gamma - 1), \tag{6}$$

in which ϵ is the internal energy, while E in the equations above was the total energy.

Both the equations for the pressure and the sound speed originate from the equation of state, while the total energy is found from the numerically evolved velocity and internal energy (eq. 7),

$$E = \rho(u^2/2 + \epsilon + \phi). \tag{7}$$

Parameter	Value
Gravitational potential	$\phi = 0.1y$
Gravitational acceleration	$g = -0.1\hat{y}$
Density	$\rho = \begin{cases} \rho = 5, 2 & \text{if y > 0,} \\ \rho = 0.5, 1 & \text{if y < 0.} \end{cases}$
Pressure	$P = P_0 - \phi\rho,\ P_0 = 2.5$
Adiabatic index	$\gamma = 7/5 = 1.4$
Velocity	$\begin{cases} v_x = 0 \\ v_y = 0.01 \cdot [1 + \cos(4\pi x)] \cdot [1 + \cos(3\pi y)]/4 \end{cases}$

Table 1: Initial parameters of the RTI simulations

2.2 Initial conditions

In this particular report, we model two instabilities, as described in the Introduction. They have slightly different initial conditions to allow for the instabilities to develop. The RTI instability is modelled in a 100 by 300 cells rectangular box, spanning x=[-0.25,0.25] and y=[-0.75,0.75]. The gravitational potential is $\phi = 0.1y$, such that the gravitational acceleration is $\vec{g} = -0.1\hat{y}$, to maintain hydrostatic equilibrium with the pressure mentioned bellow. Density is set up such that the "top" fluid (above y=0) is at $\rho = 2$ in one simulation and $\rho = 5$ in the other, and the "bottom" fluid (bellow y=0) is $\rho = 1$ in one simulation and $\rho = 0.5$ in the other. The difference in density between the two fluids can be expressed through the Atwood number. This is $\frac{\rho_1 - \rho_2}{\rho_1 + \rho_2}$, which give $A = 0.3$ in the first simulation and $A = 0.8$ in the second. $P = P_0 - \phi\rho$, where $P_0 = 2.5$. Finally, the adiabatic index is $\gamma = 7/5 = 1.4$, which corresponds to a diatomic gas. We add a small starting velocity in the y direction $v_y = 0.1 \cdot [1 + \cos(4\pi x)] \cdot [1 + \cos(3\pi y)]/4$ to help perturbate the fluids. This is summarised in table 1. The initial conditions attempted for RMI are similar, but the gravitational force is dampened to -0.01, and the initial velocity is set as $v_y = 0.5 \cdot \max(c_s)$ at $\min(y)$ and 0 everywhere else.

The KHI is modelled in a 216 by 216 cells square box, spanning 0 to 1 in both dimensions, with no gravitational potential nor acceleration. This is important in part because KH instabilities are dampened or even stabilised by gravitational acceleration (Zhou et al., 2021). The adiabatic index is $\gamma = 5/3$, which corresponds to a monoatomic gas, and the pressure constant everywhere at $P = 2.5$. Both the density (eq. 8),

$$\rho(x, y) = \begin{cases} \rho_1 - \rho_m \exp((y - 1/4) \cdot L) & \text{if } 0 \leq y \leq 1/4 \\ \rho_2 + \rho_m \exp((-y + 1/4) \cdot L) & \text{if } 1/4 \leq y \leq 1/2 \\ \rho_2 + \rho_m \exp(-(3/4 - y) \cdot L) & \text{if } 1/2 \leq y \leq 3/4 \\ \rho_1 - \rho_m \exp(-(y - 3/4) \cdot L) & \text{if } 3/4 \leq y \leq 1 \end{cases} \tag{8}$$

where $\rho_1 = 1$, $\rho_2 = 2$ and $\rho_m = (\rho_1 - \rho_2)/2$, with $L = 0.025$, and the velocity (eq. 9 and 10),

$$v_x(x, y) = \begin{cases} U_1 - U_m \exp((y - 1/4)/L) & \text{if } 0 \leq y \leq 1/4 \\ U_2 + U_m \exp((-y + 1/4)/L) & \text{if } 1/4 \leq y \leq 1/2 \\ U_2 + U_m \exp(-(3/4 - y)/L) & \text{if } 1/2 \leq y \leq 3/4 \\ U_1 - U_m \exp(-(y - 3/4)/L) & \text{if } 3/4 \leq y \leq 1 \end{cases} \tag{9}$$

where $U_1 = 0.5$, $U_2 = -0.5$, $U_m = (U_1 - U_2)/2$ and $L = 0.025$,

Parameter	Value
Gravitational potential Gravitational acceleration	$\phi = 0$ $g = 0$
Density	$\rho(x,y) = \begin{cases} \rho_1 - \rho_m \exp((y-1/4)\cdot L) & \text{if } 0 \leq y \leq 1/4 \\ \rho_2 + \rho_m \exp((-y+1/4)\cdot L) & \text{if } 1/4 \leq y \leq 1/2 \\ \rho_2 + \rho_m \exp(-(3/4-y)\cdot L) & \text{if } 1/2 \leq y \leq 3/4 \\ \rho_1 - \rho_m \exp(-(y-3/4)\cdot L) & \text{if } 3/4 \leq y \leq 1 \end{cases}$
Pressure Adiabatic index	$P = 2.5$ $\gamma = 5/3$
Velocity	$v_x(x,y) = \begin{cases} U_1 - U_m \exp((y-1/4)/L) & \text{if } 0 \leq y \leq 1/4 \\ U_2 + U_m \exp((-y+1/4)/L) & \text{if } 1/4 \leq y \leq 1/2 \\ U_2 + U_m \exp(-(3/4-y)/L) & \text{if } 1/2 \leq y \leq 3/4 \\ U_1 - U_m \exp(-(y-3/4)/L) & \text{if } 3/4 \leq y \leq 1 \end{cases}$ $v_y(x,y) = 0.01sin(4\pi x)$

Table 2: Initial parameters of the KHI simulation

$$v_y(x,y) = 0.01sin(4\pi x), \tag{10}$$

have much more complex initialisation than for the RTI simulation. These parameters are summarised in table 2. At high resolution, the combination of density and velocity disparity between the two fluids should produce RT instabilities within the KH vortrices (Matsumoto & Hoshino, 2004).

All the initial conditions used for this paper are taken from O'Connor (2026), except those for the RM instabilities.

2.3 Numerical methods

During the simulation and after initialisation, the first step is to find the primitive variables, ρ, u_j, P, ϵ and ϕ at the boundaries between the cells. This is done using the minmod method, by calculating two slopes for each cells and each dimensions, $(\rho_{m+1}-\rho_m)/\Delta x$ and $(\rho_m-\rho_{m-1})/\Delta x$, and for each variables (here shown with ρ and x). If they both have the same sign, then the slope closest to 0 is used. If they differ, then the slope is set to 0, such that the variable at the boundary is equal to the variable within the cell. The resulting variable at the boundary is the variable in the cell plus 0.5 times the slope. The density, velocity and internal energy are found this way, while the pressure, the sound speed and the total energy are derived from these variables, and ϕ does not change from one time step to the next. Using the primitives, we can calculate the fluxes through each boundaries for each cell. These fluxes, $\mathbf{F}(\mathbf{U})$ in the compact form of the Euler equation, are $\mathbf{F_i}(\mathbf{U}) = [\rho u_i, \rho u_i u_j + \delta_{ij} P, (E+P)u_i]$. However, because of the way the variables at the boundaries are found, there can be a discrepancy between the values on each sides of a same boundary, as they are found from different cells. We use the HLLE approximation (eq. 11) to solve this problem, called the Riemann problem.

$$[\mathbf{F_x}(\mathbf{U})]^{(n)}_{m+1/2} = \frac{\lambda_R \mathbf{F_x}(\mathbf{U})(p^{L,n}_{m+1/2}) - \lambda_L \mathbf{F_x}(\mathbf{U})(p^{R,n}_{m+1/2}) + \lambda_R\lambda_L(\mathbf{U}(p^{R,n}_{m+1/2}) - \mathbf{U}(p^{L,n}_{m+1/2}))}{\lambda_R - \lambda_L} \tag{11}$$

Where λ_R and λ_L are the wave speeds, taken as $\lambda_R = \max(0, u_R + c_{s,R}, u_L + c_{s,L})$ and $\lambda_L = \min(0, u_R + c_{s,R}, u_L + c_{s,L})$. We cap the difference of $\lambda_R - \lambda_L$ at 10^{-12} to make sure that small

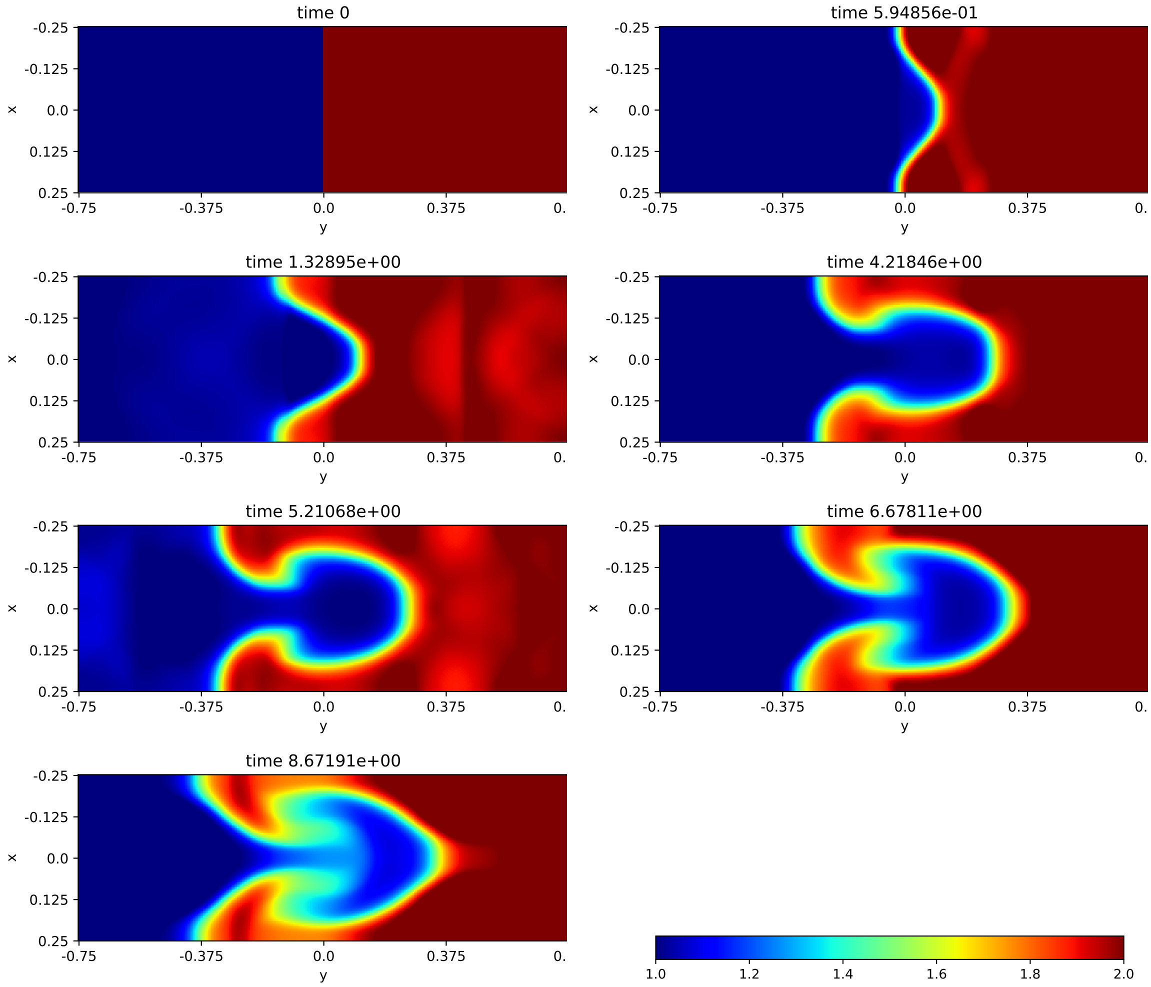

Figure 1: Density within the Rayleigh-Taylor instability simulation with an Atwood number of 0.33 at seven points in time.

numerical errors which could round it to zero do not break the model.

Finally, the updated conserved variables $\mathbf{U}$ can be found from these fluxes and the additional forces $\mathbf{S}(\mathbf{U})$ (see Hydrodynamics). This is equation 12,

$$U_m^{(n+1)} = [-(\frac{[\mathbf{F_x}(\mathbf{U})]_{m+1/2}^{(n)} - [\mathbf{F_x}(\mathbf{U})]_{m-1/2}^{(n)}}{\Delta x} + \frac{[\mathbf{F_y}(\mathbf{U})]_{m+1/2}^{(n)} - [\mathbf{F_y}(\mathbf{U})]_{m-1/2}^{(n)}}{\Delta y}) + \mathbf{S}(\mathbf{U_m^{(n)}})] \cdot \Delta t + U_m^{(n)}. \tag{12}$$

We also use a dynamical timestep wich depends mainly on the velocity and sound speed within the cells and the resolution: $\Delta t = C\frac{\min(\Delta x, \Delta y)}{\max(|u+c_s)} \cdot D$, with D the number of dimensions and $C = 0.25$. The accuracy of the evolution in time is improved by using a very basic second order Runge-Kutta method. The variables are first found after half a time step, before being reinserted into the program to find the fluxes at the full time step from them. The variables after a full time step are found by updating the variables from the previous step using the fluxes from this half-step.

Finally, the boundaries of the system are handled in the same way for all simulations, as periodic in x and reflective in y, using one row of ghost cells.

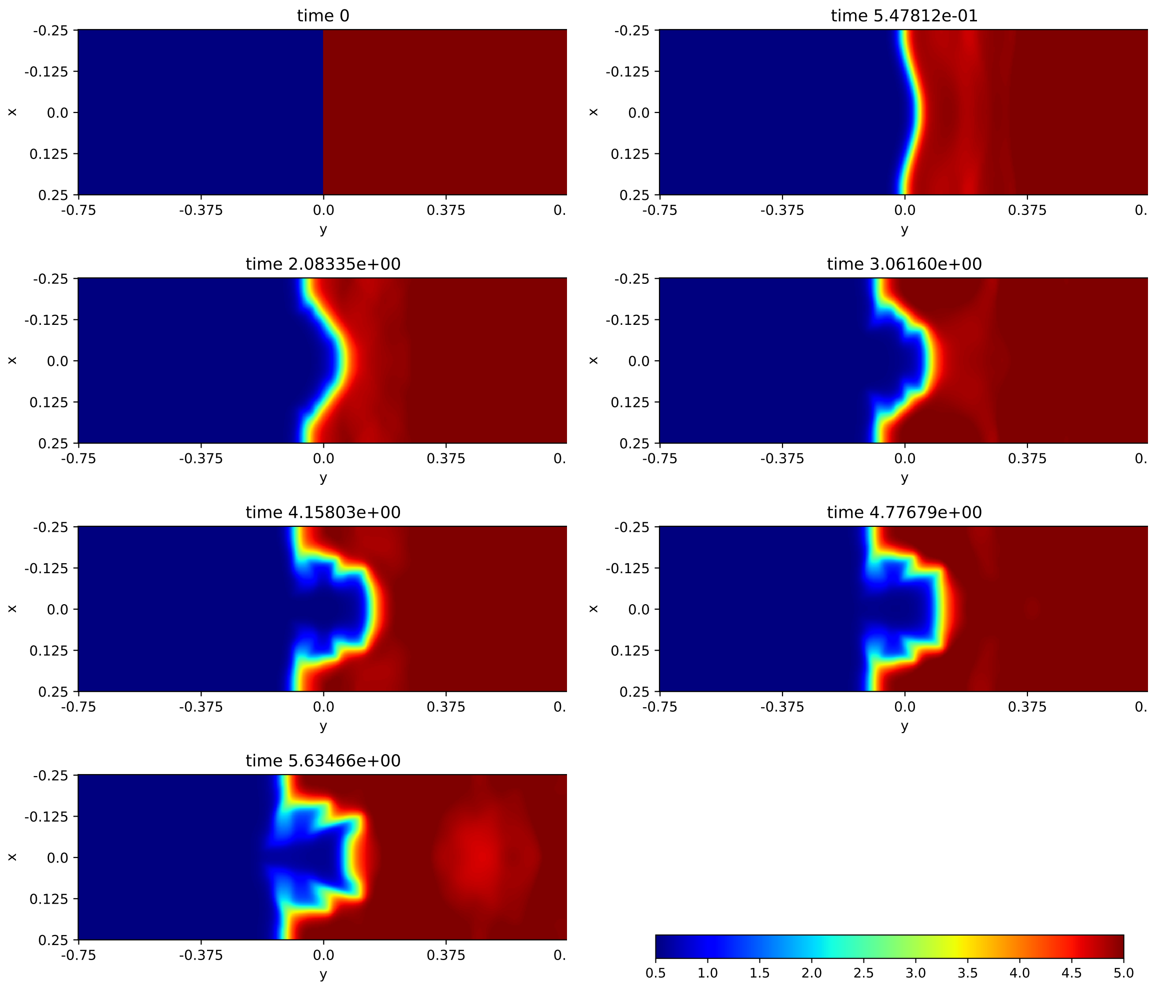

Figure 2: Density within the Rayleigh-Taylor instability simulation with an Atwood number of 0.8 at six points in time.

3 Results

First the results from the RT instability simulations. In figure 1, we can see the development of a RT instability across about 8.5 seconds, with an Atwood number $A = 0.33$. For this difference between the densities on each sides of the interface, it is expected that a finger-like shape should form towards the denser fluid, before developing into a mushroom-like shape with smaller roll-ups on each sides (Zhou et al., 2021). The instability does grow as expected, although the interfaces are more diffuse than they should be, and the mushroom shape seems to start expanding in a second even more diffuse finger in the last snapshot.

In figure 2, we can see the development of a RT instability across about 5.5 seconds, with an Atwood number $A = 0.8$. As one of the fluids is significantly denser than the other, it is expected that the less dense one should expand in bubble-like shapes toward the denser (Zhou et al., 2021). This also fit with what we can observe in the snapshots, although the bubbles seem to grow with much sharper angles than is expected from most natural phenomena.

Finally, figure 3 shows snapshots of the simulation of the RM instability with $A = 0.3$. This instability should develop similarly to an RT instability, possibly at a different rate (Zhou et al., 2021). That is not what we can observe here. It seems that the interface is instead diffusing into a larger, smoother zone.

The results from the KH instability simulation is shown in figure 4 show six snapshots from the simulation of the KH instability. The layers moving relative to each others should

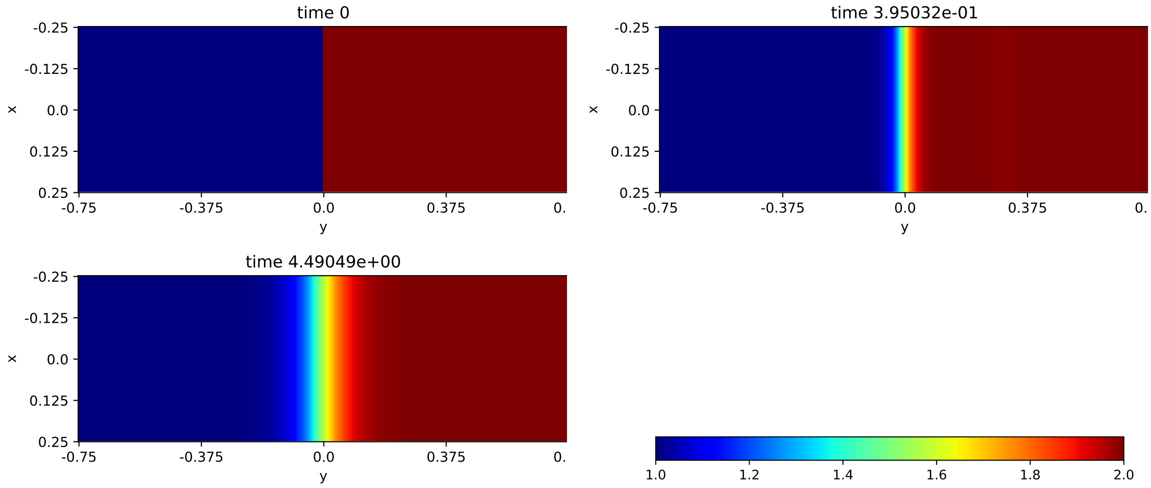

Figure 3: Density within the Richtmyer-Meshkov instability simulation at three points in time.

produce waves which keep rolling onto themselves before flattening slightly, which is exactly what we can see here.

4 Discussion

While the model itself is working and produces basic simulations of various instabilities, there are still some points which could be improved upon. First, the resolution of the model is extremely important for the instabilities to develop correctly. As can be seen in the top panel of figure 5, if the resolution is not high enough, the rolls characteristic of KH instabilities smear instead of staying sharp. If the cells are too large, the sharp interfaces are diffused across multiple cells, and the perturbation is lost within the width of the interface. The bottom panel of figure 5 shows a time ulterior to those shown in Results for the RT instability simulation where A=0.3. We can see that instead of the characteristic mushroom shape and multiple roll-ups there should be at this time, the interfaces are no longer sharp and the fluids mix. The sharper angles than expected in figure 2 could also be due to this resolution problem. The KH instability simulation which is at a much higher resolution develops exactly as expected. It is probable that given the time and the computational power, a higher resolution simulation of both RT instability simulations would produce similar results.

Second, the implementation of the Richtmyer-Meshkov instability did not go as planned. The instability should resemble Rayleigh-Taylor instabilities (Zhou et al., 2021), but presented instead as the interface becoming more diffuse. This is probably not due to a mistake within the model itself but in the initialisation. The velocity of the shock-wave might not have been sufficient, and the material it crossed before reaching the interface probably slowed it too much to have an effect. In conjonction with this, the resolution had to be lowered due to computational cost. As mentioned above, small initial perturbations could have been lost within the size of the cells and prevented from growing.

Finally, while both RTI and KHI implementations worked very well for their resolution, showing the characteristics expected for these instabilities and initial parameters, some elements could be added in further implementations. In particular, magnetic fields play an important roles in many phenomena which involve these instabilities, such as in Earth's ionosphere for RT instabilities (Zhou et al., 2021), and the Earth's magnetosphere for KH instabilities (Matsumoto & Hoshino, 2004). It would not be extremely complex to add a term representing Lorentz force to the Euler equations to account for magneto-hydrodynamics. Magnetic fields

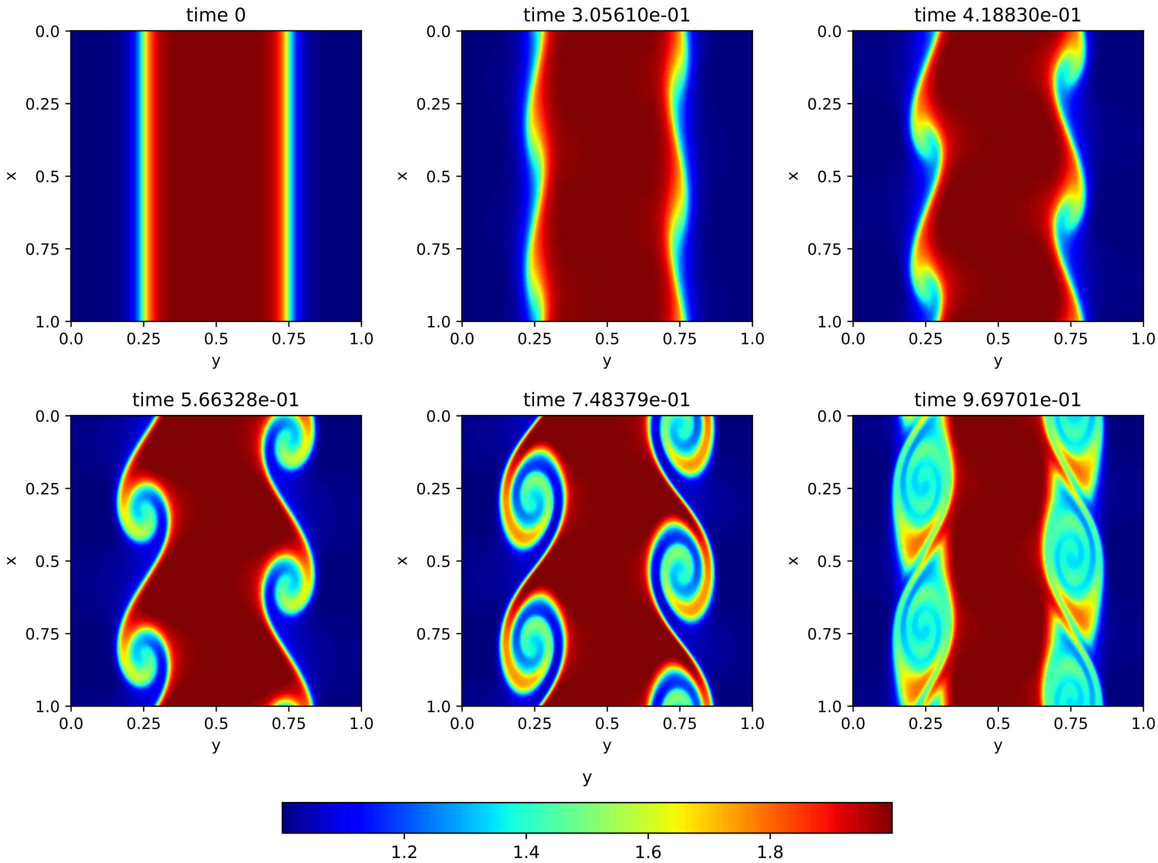

Figure 4: Density within the Kelvin-Helmholtz instability simulation at six points in time.

perpendicular to the flow are expected not to have an impact on the instabilities, while those parallel to the flow should dampen them (Trussoni, 2008).

The scale of the simulations could also be an interesting parameter to modify. At larger scales and higher resolutions, it should be possible to observe further developments within the instabilities, although higher order solutions might be needed. At smaller scales, it should be possible to observe multiple perturbations developing at the same time, as is often the case in real-world phenomena.

5 Conclusion

This report produced a basic working model for numerical simulations of a few instabilities, and showed its results for both Rayleigh-Taylor and Kelvin-Helmholtz instabilities. While Richtmyer-Meshkov instabilities did not develop as expected, this seems to be linked to initial conditions more than the model itself. The resolution of the simulations seem to have a large impact on the results, lower resolutions having a tendency to diffuse interfaces which should remain sharp. The model could also easily be developed further by adding effects such as magnetic fields, which are important in many real-world instances of these instabilities.

The code, a LATex version of this report and all the plots produced during the simulations can be found here.

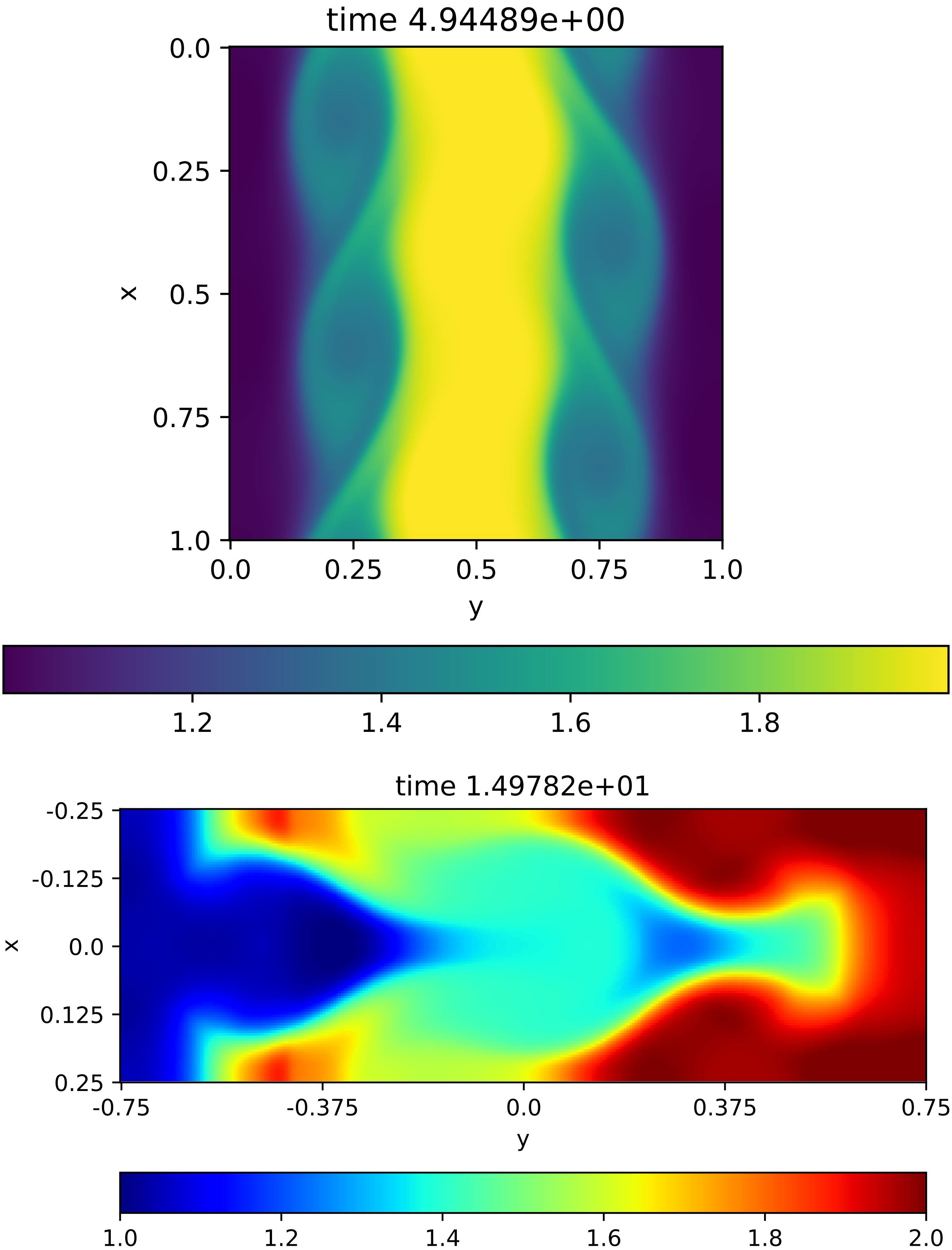

Figure 5: Top: Kelvin-Helmholtz instability after nearly 5s, with a 256 by 256 resolution. Bottom: Rayleigh-Taylor instability after nearly 15s, with a 100 by 300 resolution. The structures which should form smeared into much more diffuse elements.

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